What makes an asset the dominant intermediary?

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

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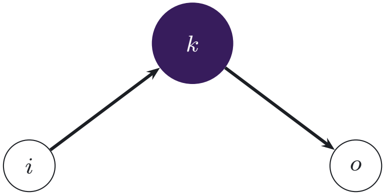
Question: what makes a vehicle currency dominant?

One asset sits between many buyers and sellers. What gives it the largest share of intermediation?

1. The cheapest asset to hold and trade
2. The asset with the deepest pairing network
3. The asset favoured by exchange architecture and routing technology
4. Different assets may dominate by trade count, value, or network position

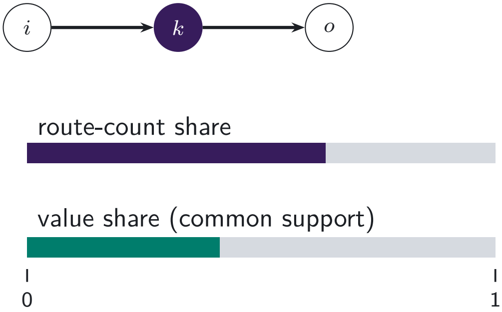


Which force matters most?

Measuring vehicle dominance

MEASUREMENT

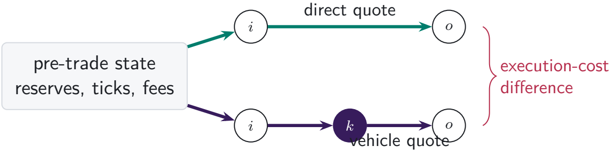
- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.



The blockchain reveals chosen routes and contemporaneous alternatives

RESEARCH DESIGN

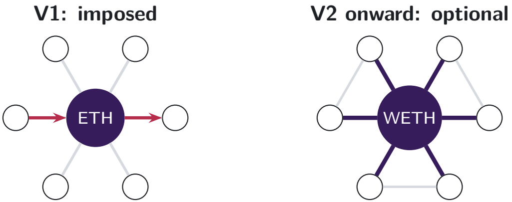
- FX data typically reveal executed trades.
- On-chain reserves and fees also let us price feasible alternatives.
- The comparison holds market conditions fixed at execution.



Protocol design first imposed ETH intermediation

INSTITUTIONAL FACT

- Uniswap V1 created one exchange contract per token, paired with ETH.
- Token-to-token swaps therefore moved through ETH by construction.
- V2 allowed arbitrary pairs; the ETH-centred pairing network persisted.



ETH remained central after arbitrary pairs became possible.

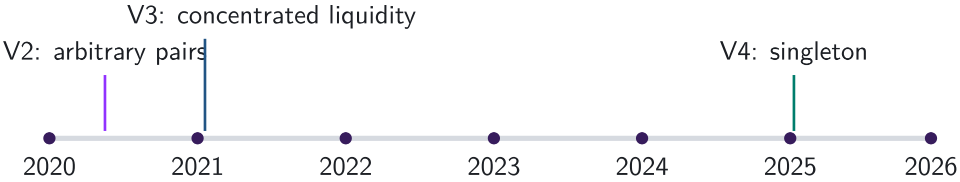
The panel follows the market across six years

DATA

2,332
calendar days

471.6m
swap legs in reconstructed routes

8
DEX deployments



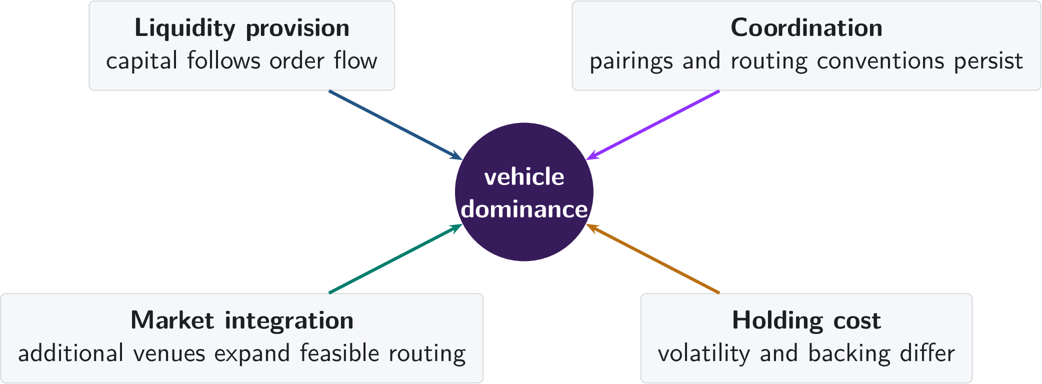
Routes are reconstructed transaction by transaction, then aggregated to daily and weekly observations.

Four measures capture use, scale, and network position

1. **Vehicle share**
   Fraction of route units intermediated by asset k.
2. **Excess use**
   Intermediary share relative to endpoint demand.
3. **Network position**
   Degree, strength, and betweenness on the trading graph.
4. **Execution advantage**
   Direct-route output relative to a feasible vehicle route at the same pre-trade state.

Together, the measures distinguish use, scale, network position, and execution cost.

Four forces can sustain vehicle dominance



The empirical design separates formation, rotation, and persistence

FORMATION

Architecture changes
Imposed intermediation, arbitrary pairing, concentrated liquidity, and singleton settlement.

ROTATION

Panel variation
Count share, value share, excess use, network position, and route complexity over time.

PERSISTENCE

Contemporaneous alternatives
An incumbent retains volume despite a cheaper feasible route.

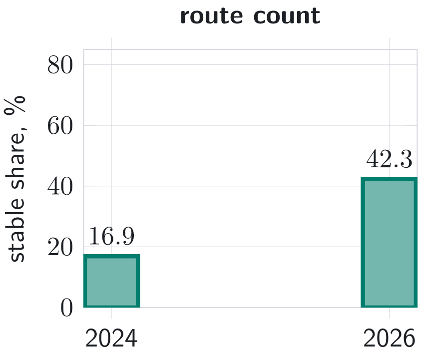
how the role begins

how its share moves

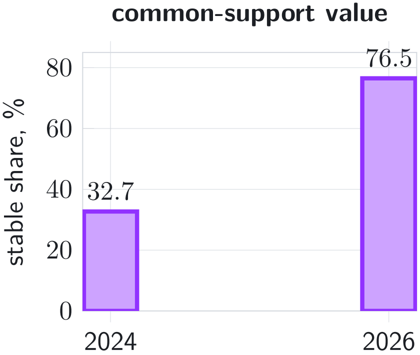
how long it survives

Stablecoins gained vehicle share by count and by value

DESCRIPTIVE RESULT



Change: **25.4 pp** SE 1.1 pp



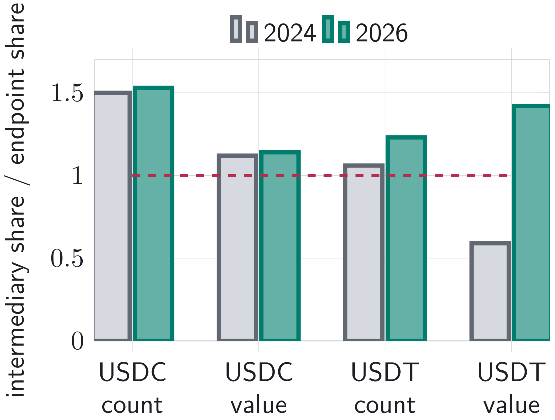
Change: **43.9 pp** SE 2.0 pp

Observed two-leg routes with comparable dollar values; daily shares are equally weighted within endpoint-year.

USDT drives the shift toward stablecoin intermediation

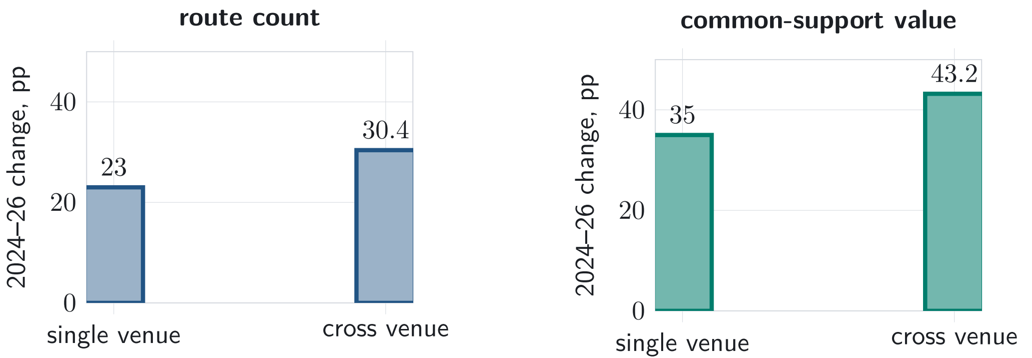
TOKEN DECOMPOSITION

- USDC was already used more as an intermediary than its endpoint share implies.
- USDT shifted from route endpoints toward intermediation.
- USDC and USDT account for 92.1% of the stable count-share increase.



Rotation is present within venues and stronger across venues

MARKET INTEGRATION



The increase is **7.5 pp** larger for cross-venue routes by count (SE 1.3) and **8.2 pp** larger by value (SE 2.3).

Dominance reflects coordination, demand, and market design

1. **A protocol can create a vehicle role.**
   V1 imposed ETH intermediation; the ETH-centred pairing network persisted after arbitrary pairs became possible.
2. **The dominant vehicle can rotate.**
   Stablecoins gained 25.4 percentage points by route count and 43.9 points by common-support value from 2024 to 2026.
3. **USDT drives the transition.**
   Its intermediary share rose above its endpoint share, and the rotation is stronger in cross-venue routing.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

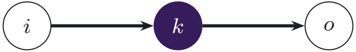
Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

A2. Backing regimes vary over time

Native platform asset
ETH, WETH; volatile settlement asset and protocol-native demand.

Fiat-reserve stablecoin

USDC, USDT; reserve and redemption design.

On-chain collateralised
DAI, LUSD, crvUSD; collateral, liquidation, and backing changes.

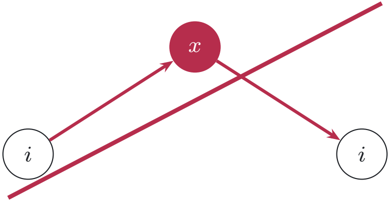
Tokenised external asset

WBTC, tokenised gold; wrapped claim on an external asset.

Heterogeneity tests use each asset's contemporaneous backing regime.

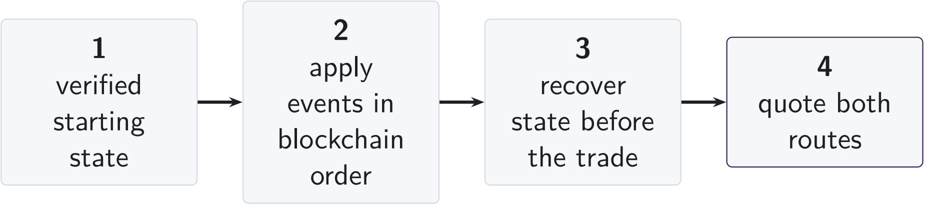
A3. Sample restrictions preserve value measurement

- Round trips are excluded from intermediation measures.
- Dollar values require consistent prices and verified token identities.
- We retain routes whose two legs imply dollar values within the pre-specified tolerance.



round trip: excluded from value-weighted vehicle measures

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. AMM designs require different state variables

Constant product
Reserves, fee, invariant. **Concentrated**

liquidity
Active liquidity, price, tick, and initialised-tick changes. **StableSwap**

families
Balances, amplification, rates, and oracle state.

Weighted pools
Vault balances, weights, scaling, and rate providers. **V4 singleton**

Pool key, tick state, and fee for standard pools without custom hooks.

Estimates include protocol families whose balances, fees, and pricing state can be independently validated.

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- The paired legs report equal ETH quantities in 87.4% of identified forced routes.
- This two-leg signature identifies protocol-imposed intermediation.

217,003
forced routes

A7. Daily and weekly frequencies answer different questions

1. **Primary panel**	day observations; date fixed effects
2. **Calendar sensitivity**	day observations; week and weekday fixed effects
3. **Aggregation robustness**	complete weeks; week fixed effects
4. **Transition series**	exact 1-, 7-, 30-, and 120-day links; HAC inference

Weekly ratios are formed from weekly sums; results are reported for weeks ending on each day of the week.

A8. Count and value define separate dominance margins

Count-weighted share

$$S_{k,t}^{N} = \frac{N_{k,t}^{I}}{\sum_j N_{j,t}^{I}}$$

Each reconstructed route carries one unit of weight.

Value-weighted share

$$S_{k,t}^{V} = \frac{\mathrm{Vol}_{k,t}^{I}}{\sum_j \mathrm{Vol}_{j,t}^{I}}$$

Each route carries reliably priced dollar value on common support.

Differences between count and value shares may reflect trade size, route structure, or the mix of intermediary assets.

A9. USDT gains intermediary use beyond endpoint demand

Route-count share

- Intermediary-minus-endpoint share rises by **2.39 pp**.
- HAC s.e.: 0.89 pp.

Dollar-value share (common support)

- Gap moves from -7.13 **pp** to $+8.14$ **pp**.
- Change: 15.27 pp; HAC s.e.: 1.50 pp.

USDT's intermediary share rises above its endpoint share.

A10. Capital, inventory, and depth are distinct quantities

Deposited capital
Funds committed to a pool; on-chain balances or independently audited TVL.

Inventory

Exact token amounts held by the pool.

Executable depth
Amount that can be traded for a given price impact under current pool state.

Quote quality

Execution price for trade size q along feasible routes.

We measure capital with TVL and execution depth from the pool's pricing function.

A11. Integration changes both reach and route choice

Available trading opportunities

- active venues
- feasible paths
- available intermediary assets

Route choice conditional on availability

- direct versus intermediated
- single- versus cross-venue
- sequential versus split routing

We test whether stablecoin intermediation rises more when additional venues and routes are available, conditional on endpoint pair and date.

A12. References: vehicle currencies and market structure

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